Instructions:

- **Due:** March 30, 11:59PM . Upload one **single PDF** to Blackboard containing your complete submission and upload all associated LaTeX and MATLAB source files.
- Each problem is worth $\frac{100}{\text{number of problems}}$ points.
- Start each problem on a new page.
- Label your PDF file exactly as

  `ENME408_S26_HW05_YourLastName.pdf`

- All work must be typeset in LaTeX. Handwritten or scanned submissions will not be accepted.
- Failure to follow these submission and formatting guidelines may result in a substantial point deduction or a score of zero.

Learning Objectives Many dynamical systems that arise in UAV control are unstable or sensitive to disturbances and therefore require feedback control for stabilization and performance. The goal of this assignment is to develop insight into how feedback can be used to stabilize such systems and improve their behavior in the presence of disturbances. In this homework, you will (i) linearize nonlinear rigid-body rotational dynamics, (ii) design a state-feedback controller to stabilize a double-integrator system, (iii) investigate the limitations of PD control when constant disturbances are present, and (iv) design a controller with integral action to achieve disturbance rejection and constant command following. Through analysis and simulation, you will observe how different controller structures influence stability, transient response, and steady-state performance.

Problem 1. Consider the rigid-body rotational dynamics of a vehicle

$$\dot{q} = S(q)^{-1}\omega, \tag{1}$$
$$J\dot{\omega} + \omega^{\times} J\omega = \tau, \tag{2}$$

where $q = \begin{bmatrix}\phi & \theta & \psi\end{bmatrix}^{\mathrm{T}}$ is the vector of Euler angles, $S(q)$ is defined in Section 7 of the GKSD notes, $\omega \in \mathbb{R}^3$ is the angular velocity expressed in the body frame, $J \in \mathbb{R}^{3\times 3}$ is the constant inertia matrix, $\tau \in \mathbb{R}^3$ is the control torque, and $\omega^{\times}$ denotes the skew-symmetric matrix associated with ω.

1. Define $x \triangleq \begin{bmatrix} q \\ \omega \end{bmatrix}$ and $u = \tau$. Write the system in the state space form $\dot{x} = f(x, u)$.
2. Determine an equilibrium point (x_e, u_e) of the nonlinear system.
3. Linearize the nonlinear dynamics about the equilibrium point obtained above to obtain the linear system $\dot{\xi} = A\xi + Bv$, where $\xi = x - x_e$ and $v = u - u_e$.

Problem 2 (Stabilization of a Double Integrator)**.** Consider the double-integrator system

$$\ddot{y} = u, \tag{3}$$

where $y \in \mathbb{R}$ is the position and $u \in \mathbb{R}$ is the control input.

1. Define the state vector $x_1 = y$, and $x_2 = \dot{y}$. Write the system in the state-space form $\dot{x} = Ax + Bu$.
2. Show that the open-loop system is not asymptotically stable.
3. Consider the state-feedback control law $u = Kx$. Derive the closed-loop system dynamics.

4. Find the characteristic polynomial of the closed-loop system and determine conditions on K that guarantee asymptotic stability.

5. Assume the desired closed-loop characteristic polynomial is $s^2 + 2\zeta\omega_\mathrm{n}s + \omega_\mathrm{n}^2$. Determine the gain K that achieves this desired characteristic polynomial. Let $\zeta = 0.7$ and $\omega_\mathrm{n} = 2$ rad/s. Compute the corresponding K and the closed-loop eigenvalues.

6. Simulate the closed-loop system with the gains obtained above and initial conditions $y(0) = 1$, and $\dot{y}(0) = 1$. Plot $x(t)$ and briefly describe the behavior of the system.

7. Now suppose that the system is affected by a constant disturbance d such that

$$\ddot{y} = u + d. \tag{4}$$

Simulate the closed-loop system with $d = 0.1$. Plot $x(t)$ and comment on the steady-state behavior. Based on your simulation results, explain why the LQR controller without an integrator cannot completely reject a constant disturbance.

Problem 3 (Disturbance Rejection and Constant Command Following)**.** Consider again the double-integrator system (4).

1. Define the tracking error $e = y - r$, where r is a constant reference position. Explain why the PD controller designed in the previous problem cannot guarantee that $e(t) \to 0$ in the presence of a constant disturbance.

2. To improve disturbance rejection and enable constant command following, introduce the integral state

$$q(t) \triangleq \int_0^t e(\tau)\, d\tau.$$

Define the augmented state vector $x_\mathrm{a} = [\; y \;\; \dot{y} \;\; q \;]^T$. Write the augmented system dynamics in the form

$$\dot{x}_\mathrm{a} = A_\mathrm{a}x_\mathrm{a} + B_\mathrm{a}u + B_{r,\mathrm{a}}r + B_{d,\mathrm{a}}d.$$

3. Consider the control law $u = k_\mathrm{p}y + k_\mathrm{d}\dot{y} + k_\mathrm{i}q = K_\mathrm{a}x_\mathrm{a}$. Derive the closed-loop dynamics of the augmented system. Show that the characteristic polynomial of the closed-loop system is $s^3 - k_\mathrm{d}s^2 - k_\mathrm{p}s + k_\mathrm{i}$.

4. Suppose the desired closed-loop characteristic polynomial is $(s^2 + 2\zeta\omega_\mathrm{n}s + \omega_\mathrm{n}^2)(s + p)$, where $p > 0$ determines the speed of the integral dynamics. Expand the polynomial and determine the gains k_p, k_d, and k_i. Let $\zeta = 0.7$, $\omega_\mathrm{n} = 2$, and $p = 1$. Compute the corresponding controller gains.

5. Simulate the closed-loop system for the reference command $r(t) = 1$ with initial conditions $y(0) = 0$, $\dot{y}(0) = 0$, and $q(0) = 0$. Plot $y(t)$ and briefly describe the tracking performance.

6. Repeat the simulation with a disturbance $d = 0.1$. Plot $y(t)$ and compare the steady-state behavior with the LQR controller from the previous problem.